



Austroads

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Austroads Road Deterioration Model Update

Cracking

Austrroads Road Deterioration Model Update: Cracking

Prepared by

Dr Tim Martin and Ranita Sen

Project Manager

Dr Zahid Hoque

Publisher

Austrroads Ltd.
Level 9, 570 George Street
Sydney NSW 2000 Australia
Phone: +61 2 8265 3300

austrroads@austrroads.com.au
www.austrroads.com.au



Abstract

The aim of the project was to update the current Austrroads road deterioration (RD) models using the long-term pavement performance and long-term pavement performance maintenance (LTPP/LTPPM) dataset and other data available, such as the traffic speed deflector (TSD) datasets, to improve these models' explanatory power.

The cracking RD model documented in this report was based on a mechanistic-empirical, deterministic approach using a multi-variate non-linear regression (MVNLR) analysis. The cracking RD model uses cumulative cracking as the dependent variable for thinly surfaced flexible unbound granular pavements. The model was developed using the New South Wales dataset collected by the TSD from 2014 to 2018. The cracking RD model will need calibration to suit locally observed cracking conditions as location has a bearing on the rates of cracking progression.

Keywords

prediction of cracking progression, flexible thin surfaced unbound granular pavements,

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- Department of State Growth Tasmania
- Department of Infrastructure, Planning and Logistics Northern Territory
- Transport Canberra and City Services Directorate, Australian Capital Territory
- Department of Infrastructure, Transport, Regional Development, Communications and the Arts
- Australian Local Government Association

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This report has been prepared for Austrroads as part of its work to promote improved Australian and New Zealand transport outcomes by providing expert technical input on road and road transport issues.

Individual road agencies will determine their response to this report following consideration of their legislative or administrative arrangements, available funding, as well as local circumstances and priorities.

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Summary

Road deterioration (RD) models help in predicting future road condition and assessing maintenance needs. The current Austroads RD models available for use by road agencies and industry were developed in 2008 using observational data collected from 1994 to 2007 on 39 Austroads long-term pavement performance (LTPP) sections and 40 Austroads long-term pavement performance maintenance (LTPPM) sections. Additional supplementary experimental data from the accelerated loading facility (ALF) was also used. A further 10 years of observational data was collected on the Austroads LTPP/LTPPM sites in the period from 2008 to 2018. Furthermore, the quality, coverage, and scope of network observational data has also been significantly extended with the introduction of the traffic speed deflectometer (TSD) in 2014. The TSD provides substantially more detailed condition and structural data collected concurrently across the state and territory arterial road networks.

The aim of the project was to update the current Austroads RD models using the additional LTPP/LTPPM data and other data available, such as the TSD datasets, to improve these models' explanatory power to support wider loading and climate conditions observed in Australia. A proof-of-concept approach based on using the available TSD datasets was undertaken early in the project to determine if these datasets could be used initially for developing a refined structural RD model with the possibility of this approach being applied to the other RD models.

The updated cracking RD model is applicable to thin surfaced unbound granular base flexible pavements undergoing gradual deterioration. The thin surfaces include both sprayed seals and thin asphalt (≤ 40 mm) surfaces.

The cracking RD model documented in this report was based on a mechanistic-empirical, deterministic approach using a multi-variate non-linear regression analysis. The cracking RD model uses cumulative cracking, $\sum \Delta crx$, as the dependent variable for thinly surfaced flexible unbound pavements. The model was developed using the New South Wales dataset collected by the TSD from 2014 to 2018. The model has two significant independent variables: cracking age, $crxAGE$, and the Thornthwaite Moisture Index, TMI , as shown in Equation 1. Although the independent variables are statistically significant, the model has an indeterminate fit (r^2) to the data. Consequently, the cracking RD model will need calibration using the calibration coefficient, K_c , to suit locally observed cracking conditions as the evidence suggests that location also has a bearing on the rates of cracking progression.

$$\sum \Delta crx = K_c \times [\text{EXP}[0.227 \times crxAGE \times (100 + TMI)/100] - 1] \quad 1$$

As the cracking RD model development was highly dependent on the accuracy of the estimation of sprayed seal life and knowledge of the sprayed seal age, further refinement of sprayed seal life models needs to be undertaken as well as being able to accurately determine the types of bitumen and additives used in the binders being assessed for cracking by the TSD. This will allow application of the most appropriate seal life models, when available, to determine with a reasonable degree of reliability the sprayed seal life and hence the cracking age.

An alternative probabilistic approach to estimating crack initiation based on multivariate logistic regression with sprayed seal samples that are uncracked and lightly cracked should also be considered. This could be in conjunction with the current Queensland Transport and Main Roads NACOE research program aimed at developing reliable sprayed seal life models.

However, further refinement of the cracking RD models should not be undertaken until several more years of TSD cracking data become available across most states and territories in conjunction with reliable sprayed seal life models.

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1. Introduction

1.1 Background

Road agencies and industry are increasingly required to predict pavement condition and make justifiable decisions using information and analyses from their pavement management systems (PMS) to:

- support future funding needs and to identify preservation and renewal candidates, accounting for the requirements of different roads and regions including road user requirements
- justify possible charges to road users, including those for heavy vehicles as part of heavy vehicle road reform (HVRR).

In each case a defensible basis is needed requiring an evidence-based understanding of how roads deteriorate over time considering traffic and environmental loading as well as maintenance practices. The full range of sealed road circumstances must also be covered, including different design standards and asset provision and condition (including surface and pavement types, shoulders, drains, etc.) with a need to incorporate significant variables in the predictive models based on their intended application. Terrain and climatic conditions are also important, with topography and subgrade soils impacting performance, as well as consideration of the contribution of long term and short term, often severe climatic patterns.

Road deterioration (RD) models help in predicting future road condition and assessing maintenance needs. The current Austroads RD models available for use by road agencies and industry were developed in 2008 (Austroads 2010a, 2010b) using observational data collected from 1994 to 2007 on 39 long-term pavement performance (LTPP) sections and 40 long-term pavement performance maintenance (LTPPM) sections, including additional supplementary experimental data (Martin 2010a, 2010b) from the accelerated loading facility (ALF).

A New Zealand LTPP program was established in 2001 with 176 sites and now has over 18 years of observational data. This program was subject to a review by Henning and Brown (2016) recommending that further information be provided to refine the current RD models used in New Zealand.

1.2 Purpose

The purpose of the project was to update the four current Austroads RD models using the additional data and information available to improve their explanatory power to support wider loading and climate conditions observed in Australia. A further 10 years of observational data was collected on the Austroads LTPP/LTPPM sites in the period from 2008 to 2018.

Furthermore, the quality, coverage and scope of network observational data has also improved significantly with the introduction of the traffic speed deflectometer (TSD) in 2014. The TSD provides substantially more detailed surface and structural condition data collected concurrently across the state arterial road networks. By incorporating this wealth of information in model development, including the TSD data, the acceptance and usefulness of the resulting more robust models for road agencies and industry was expected to be much greater.

1.3 Scope

The four RD models updated represent flexible pavements, i.e. sprayed seal and thin asphalt on unbound granular pavements. These RD models are:

1. cracking deterioration model
2. rutting deterioration model
3. roughness deterioration model
4. structural (deflection) deterioration model.

1.4 Methodology

The approach to updating the RD models included a literature review to gain state-of-art approaches to RD modelling, presentation of modelling concepts, workshops, and consultation with the project working group throughout the project duration.

The methodology adopted evolved as the project progressed based on the availability, quality and duration of the data collected. Section 4 details the methodology adopted for the cracking RD model.

2. Literature Review

2.1 Overview

A literature review was undertaken to gain a state-of-the-art appreciation of current practice in RD model development and refinement to ensure that the updating of the current Austroads RD models was undertaken using best practice. These RD models were based on a mechanistic-empirical deterministic approach using the observational and experimental data available in 2008.

The review included an analysis of the potential for feasible RD model development using the vast amount of data gained from the TSD measurement of road conditions, regarded as 'big data', over several years in New South Wales and Queensland, and more recently in Western Australia. The TSD data was collected every year for five to six years involving 17 000–20 000 km of road length in New South Wales and Queensland, respectively. Western Australia and New Zealand have also had TSD network level surveys. In 2013 the National Research Council (2013) considered model building with big data to be relatively uncharted territory.

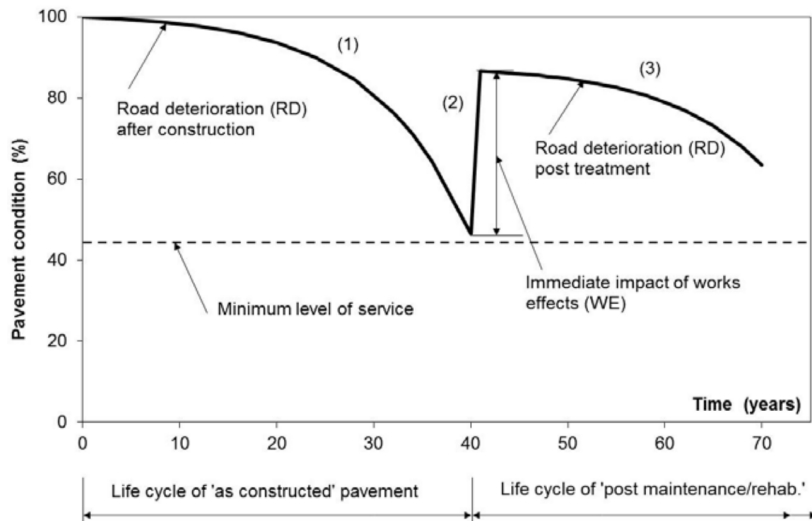
The review included a proof-of-concept investigation of the machine learning, or artificial intelligence (AI), approach made possible with the TSD data. The expectation was that 'big data' analytics applied to the TSD datasets may prove useful in identifying contributing factors to pavement deterioration and condition and guide the selection of the independent variables to be included in the RD model update. However, it was considered that it was not advisable to combine the TSD data with the relatively small LTPP/LTPPM dataset.

2.2 Pavement Performance

There are two main stages of in-service pavement performance for which most RD models apply (Martin & Choummanivong 2018). These are:

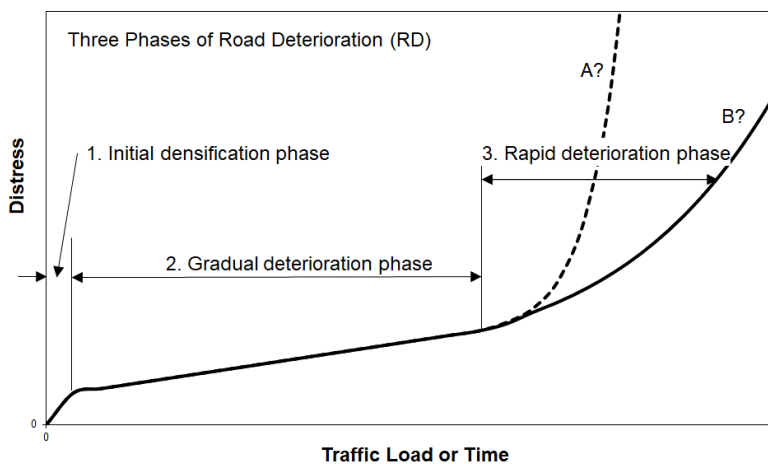
- RD of pavement condition, both functional and structural, immediately post construction. See Stage (1) in Figure 2.1
- RD of pavement conditions, both functional and structural, post works effects. See Stage (3) in Figure 2.1.

Stage (2) in Figure 2.1 is the impact of works effects (WE) which restore pavement conditions closer to what they were after original construction, although the actual effect will differ depending on treatment applied. It is usually assumed in practice that the same RD models apply to Stages (1) and (3), although more sophisticated modelling may be able to account for changes in structural conditions due to the WE which can influence the predicted rate of deterioration.

Figure 2.1: Three stages of pavement performance

Source: Martin and Choummanivong (2018).

Figure 2.2 shows the three phases of road deterioration, or distress, with the passage of time and traffic load. The first phase, initial densification, occurs within the first 12 months post construction with traffic loading. Initial densification is apparent in certain forms of permanent distress such as rutting and roughness. Initial densification may also lead to an increase in pavement strength, but this is not usually accounted for in deterministic RD modelling.

Figure 2.2: Three phases of road deterioration (RD)

Source: Martin and Choummanivong (2018).

The second phase of road deterioration, gradual deterioration, is the phase in which most in-service pavements operate. As Figure 2.2 shows, gradual deterioration proceeds at a low rate and is virtually linear. Most of the documented Austroads RD models (Austroads 2010a, 2010b) are confined to predictions in the gradual deterioration phase.

The rapid deterioration phase is the third and final phase of deterioration. The rapid deterioration phase, defined as a deterioration rate two to three times that of the gradual deterioration rate, is shown in Figure 2.2. Rapid deterioration is difficult to predict as the pavement approaches ultimate failure. Some RD models were developed for this phase (Austroads 2014, 2015); however, they are currently not fully reliable predictors of rapid deterioration due to the limited observational data.

2.2.1 Limit of Gradual Deterioration

The three phases of deterioration were simulated by ALF testing of various surface maintenance treatments and formed the basis for defining the frontier limiting the gradual deterioration phase of sealed granular pavements. This frontier was defined using a binary logistic regression analysis (Mendenhall & Sincich 1996) of the total of 71 samples identified undergoing either gradual or rapid deterioration (Martin 2008). The samples identified as undergoing rapid deterioration were defined as those deteriorating at a rate of two to three times the samples undergoing gradual deterioration.

Figure 2.3 plots the 53 samples defined as experiencing gradual deterioration as open squares and the 18 samples defined as experiencing rapid deterioration as filled squares. The frontier between gradual and rapid deterioration was defined as in Equation 2.

$$rut_{\max} = 86.347 - 11.008 \times IRI \quad 2$$

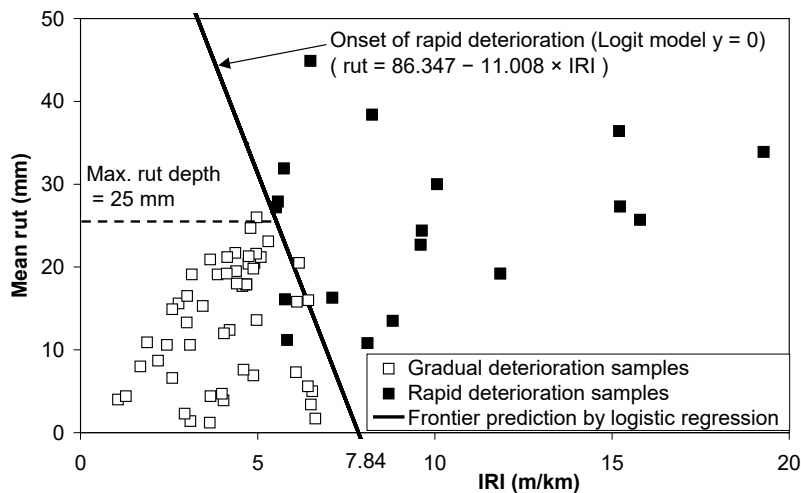
where

$rut_{\max}$ = mean maximum vertical deformation from the original surface profile (mm) with an absolute maximum value of 25 mm

IRI = International Roughness Index (m/km)

All the coefficients and the constant term in Equation 2 had relatively low values of standard error (SE) all with significance at the 95% confidence level. Equation 2 predicted the correct state of the gradual and rapid deterioration for over 94% of the total of 71 samples.

Figure 2.3: Mean rut (mm) vs roughness (IRI) for ALF maintenance experiments



Source: Martin (2008).

2.3 Assessment of Pavement Performance

2.3.1 Functional Performance Measures

Functional performance measures such as road roughness, rutting and cracking are used as measures, or indicators, of long-term pavement performance. Road roughness, as a functional performance measure, has the distinct advantage of being objectively measured by a standardised measuring device, unlike some pavement performance indices. A commonly used measure of roughness, the International Roughness Index (IRI in m/km), is defined by a polynomial function of the root mean square deviation (RMSD) from a straight-line profile measured at given intervals along a given base length (Sayers, Gillespie & Queiroz 1986).

2.3.2 Structural Indicators

Some form of structural performance indicator is needed to assess this aspect of pavement performance. While visible surface distress is a useful indicator of surface functional condition, it does not give any direct indication of structural condition. For example, surface cracking is indistinguishable from full depth structural cracking and sub-surface cracking is not visible from the surface (Eijbersen and van Zwieten 1998).

Structural capacity can be reduced either gradually or rapidly by the unplanned increased passage of traffic load (equivalent standard axles, ESAs) over time and rapidly by local shear failure of the pavement from increased axle loads. Road agencies have also increasingly decided to include the structural condition of pavements as a performance indicator for reporting asset condition, and in specific cases for monitoring contractor managed pavements on behalf of the road agency.

2.4 Review of Recent Modelling Practice

As the current project is aimed at updating the existing Austroads road deterioration models developed in 2008, the literature review covered work done on the development, refinement, calibration, and validation of flexible pavement deterioration models post 2006.

2.4.1 Typical Modelling Approaches

The following approaches are normally used to predict pavement performance (Haas, Hudson & Zaniewski 1994, Lytton 1987, Mahoney 1990):

- Probabilistic, or trend, approaches that inherently recognise the stochastic nature of pavement performance by predicting the variability of the dependent variable for both structural and functional distress deterioration
 - probabilistic approaches include Markov chain models (Haas, Hudson & Zaniewski 1994), semi-Markov chain models (Lytton 1987), Bayesian models (National Research Council 2013), logistic regression models (Austroads 2012) and fuzzy logic models (Jooste et al. 2010).
- Deterministic approaches that predict a single value of the dependent variable from pavement performance prediction models are based on statistical relationships between the dependent and independent pavement performance indicators
 - deterministic approaches include mechanistic models (Rauhut & Gendell 1987), mechanistic-empirical and empirical models (Lytton 1987), and the artificial neural networks (ANN) approach (Abdelaziz et al. 2020).

2.4.2 Machine Learning with Big Datasets

Machine learning techniques typically require very large datasets to train models or algorithms that can be used for prediction. These algorithms can provide robust models although they can also be prone to overfitting where a model will fit the training data too closely and not be transferrable to other data. The other challenge with machine learning models is interpretability of the models. While machine learning black-box models are frequently successful in producing robust models, there can be a mistrust of the results (Gilpin, Bau, Yuan, Bajwa, Specter & Kagal 2019) or they require a complex implementation.

Clustering

Clustering is an activity that partitions big data into groups so that data items within each group are similar to each other and items across different groups are not similar. K-means and hierarchical clustering are popular algorithmic approaches. Mixture models use a probabilistic framework to model clusters (National Research Council 2013).

Big data is highly amenable to the data analysis algorithms used in machine learning and statistical learning. A k-means algorithm is a useful method for the clustering of data into groups that is not based on a statistical model of the data. Optimisation formulations are needed to find the inferences from the data and determine whether the data can be clustered into coherent groups.

Machine learning (ML) models

Three main classes of machine learning models are presented in Simeone (2018), which are:

1. supervised learning, where the training data has a known target or output variable
2. unsupervised learning, where there is no output variable, and the data analysis aims to find patterns and structures in the data
3. reinforcement learning where the model receives feedback on decisions rather than having a clearly known output.

2.4.3 Application of ML to Pavement Deterioration Modelling

Tabatabae, Shafahi & Ziyadi (2013) developed a two-stage model for predicting pavement deterioration using data from the Minnesota Department of Transportation hot mix asphalt test pavement between 1993 and 2008. The model first classifies the data into like groups or families (clusters) using a support vector machine (SVM) model (a supervised ML model with associated learning algorithms that analyse data for classification and regression analysis). A recurrent neural networks (RNN) artificial intelligence (AI) approach is then used to model the deterioration. RNN models can handle time-series data (Choi & Do 2019).

2.5 Selection of Appropriate Modelling Approaches

The literature revealed the successful application of probabilistic and deterministic approaches in pavement performance modelling with varying degrees of complexity and limitations due to the availability of suitable data. Models generated through the probabilistic approach generally provide higher prediction accuracy over the deterministic models. However, deterministic equation-based models are relatively straightforward showing the interaction between predictor parameters and modelled output and are more transferable than probabilistic models due to their incorporation of multiple explanatory variables.

The decision on the selection of an appropriate modelling approach is contingent upon its intended application. The choice of modelling methodology depends on the following:

- availability of sufficient high-quality data
- preference of the road agencies (e.g. equation-based model versus ANN-based model)

- knowledge and expertise on a particular domain (e.g. machine learning, deep learning, etc.)
- the complexity of the implementation within a PMS system
- cost of implementation.

Currently most Austroads members use deterministic RD models. The reasons for this choice include:

- In-service pavement deterioration occurs in the gradual deterioration phase, which is where deterministic model predictions are likely to be most stable.
- RD models can be easily incorporated into a PMS.
- RD models can provide a relationship to traffic data which is important to consider with rising traffic volumes.
- The outputs of these models can be calibrated to reflect observed pavement performance under various loading and environmental conditions, and service level requirements.

2.5.1 Expected Modelling Approach

In this context of updating the existing RD models using the high-quality LTPP/LTPPM dataset it was expected that the current mechanistic-empirical deterministic approach will be used.

Kadar et al. (2015) have shown, on a proof-of-concept basis, that a deterministic RD model in combination with an observationally based probability density function (PDF) can be used to give both single value and probabilistic estimates of pavement conditions over time.

2.5.2 Proof of Concept for TSD Datasets

A proof-of-concept approach based on using the available TSD datasets was undertaken to determine if this approach could be used initially for developing a structural RD model with the possibility of this approach being used for the other RD models.

The independent variables currently not included in the existing structural RD model such as surface distresses, subgrade reactivity, terrain category and drainage conditions were considered for inclusion in the model in the proof-of-concept approach. The availability of time series TSD data along with the required explanatory variables (climate, drainage, soil classification etc.) determined the selection of network data that was used for the development of a proof-of-concept structural RD model.

3. Data Review

3.1 Data for Analysis

A review of the supplied historical condition and data received from road agencies and the LTPP/LTPPM datasets from Australia and New Zealand was undertaken. Data was reviewed to check the completeness of the supplied datasets with the inclusion and availability of the following additional information that potentially provided additional explanatory independent variables for updating the current RD models:

- time series TSD data
- pavement inventory information including terrain and drainage conditions
- climate information
- maintenance information.

The project also used a TSD dataset from Transport for NSW (TfNSW) in a proof-of-concept approach to determine its usefulness in developing a refined structural RD model taking into consideration the independent variables currently not included such as: surface distresses, subgrade reactivity, terrain category and drainage conditions. The structural RD model was seen as a critical element in the RD model update as it is an input variable for both the rutting and roughness RD models. The structural RD model may also be an input variable for the RD models for cracking in some cases. The Austroads LTPP/LTPPM datasets were expected to be primarily used to update the current RD models due to their completeness, as noted in Section 3.5.2. However, the cracking data in the TSD dataset from TfNSW was used to develop the cracking RD model.

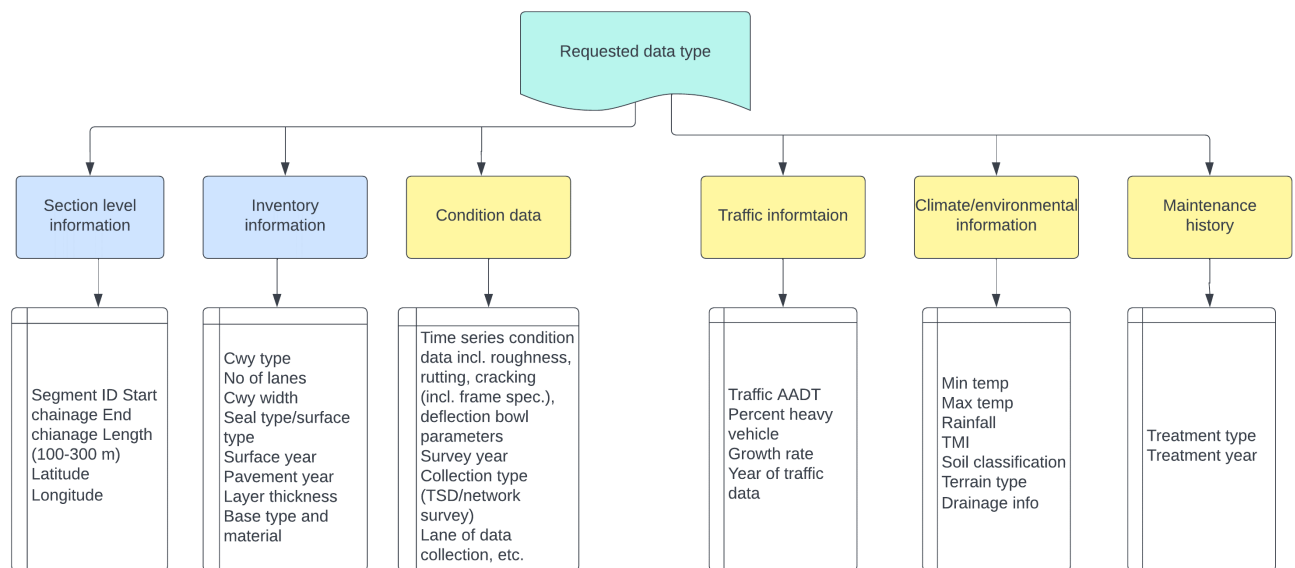
The LTPP/LTPPM and the New Zealand Transport Agency (NZTA) LTPP datasets contained most of the additional independent variables being considered for the RD model upgrade. The state and territory road agencies did not have network level deflection data, except NSW and Queensland with TSD datasets. The LTPP/LTPPM and NZTA LTPP datasets were not combined with the TSD deflection datasets because the 'big data' TSD datasets would overwhelm the LTPP datasets, as further discussed in Section 3.5.3. The other problem with combining the deflection datasets of different sizes is the difference in the length of the time series of the condition data: the NSW TSD dataset has a time series of five years while the LTPP/LTPPM datasets had a time series ranging from 18 to 24 years.

3.2 Data Requested from Road Agencies

A data template prepared by the Australian Road Research Board (ARRB) was also distributed to the road agencies for supply of the required information. Figure 3.1 details the list of the requested data types which included the following:

- section level information
- inventory information
- condition data
- traffic information
- climate/environmental information
- maintenance history.

Figure 3.1: Requested data types



3.3 Review of the Data from Australian Road Agencies

The supplied dataset from each road agency was reviewed separately to assess the completeness of the dataset as well as its coverage over time. The following sections summarise the findings from each road agency dataset.

3.3.1 Transport for New South Wales (TfNSW) Data

For the data supplied by TfNSW, surface condition information was available from 1999 onwards. The TSD data was collected every year for five years involving 15 000–20 000 km of road length in New South Wales. Consequently, deflection data was available from 2014 to 2018. This TSD data included cracking data which was subsequently used for the cracking RD model development as discussed in Section 4.1.

The TSD data collected was 1 800 km road length with time series data for three years, 9 000 km with time series data for four years and 4 000 km with time series data for five years covering 11% of the arterial road network.

3.3.2 Queensland Department of Transport and Main Roads (TMR) Data

For the data supplied by TMR, surface condition information was available from 2015 to 2020. The TSD data was collected every year for five years involving approximately 15 000 km of road length in Queensland. Therefore, deflection data was available from 2015 to 2020.

The TSD data collected around 5 000 km road length with time series data for three years, 3 400 km with time series data for four years and 7 800 km with time series data for five years covering 22% of the arterial road network.

3.3.3 Main Roads Western Australia (MRWA) Data

Surface condition data was supplied by MRWA from 2009 onwards. However, TSD deflection and cracking data was only available for 2019 (2019 data was collected over 2018 to 2019). Falling weight deflectometer (FWD) data was also supplied with a partial collection from 2002 to 2017.

MRWA 2019 TSD data was collected for the full network (17 000 km) over 2018 to 2019.

3.3.4 Department of Planning, Transport and Infrastructure (DPTI) South Australia Data

DPTI supplied surface condition data (roughness and rutting) from 2009 to 2020. Cracking data was only available over the period from 2017 to 2020 with two years of data collected. The DPTI did not have any TSD data collected, and no FWD deflection data was supplied.

3.3.5 Department of Infrastructure, Planning and Logistics (DIPL) Northern Territory Data

DIPL supplied surface condition data (roughness, rutting, cracking) for two years (2015, 2019). DIPL also had TSD deflection data collected for the whole network in 2019 which was supplied.

3.3.6 Department of State Growth (DSG) Tasmania Data

Surface condition data (roughness, rutting and cracking) supplied by DSG included for three years (2012, 2015, 2018) of data. TSD deflection data was also collected for part of the network in 2018.

3.3.7 Summary of Road Agency Data Supplied

Findings from the review of the road agency data supplied were as follows:

- Most road agencies had time series surface condition information (roughness, rutting and cracking).
- A lack of complete network-wide time series deflection data required for the structural model except for TfNSW and TMR.
- A lack of important ancillary information e.g. maintenance history, drainage, climate, etc.

3.4 Missing Data for RD Model Development

The following missing independent variables were estimated initially for the proof-of-concept approach and then later used as additional explanatory independent variables for refinement of the roughness, rutting and cracking RD models.

3.4.1 Climate Information

Historical climate information (monthly minimum, maximum and average temperature, and monthly rainfall information for years 1994 to 2018) was supplied from the Bureau of Meteorology website (BOM 2021) for:

- the weather stations close to the LTPP/LTPPM
- for 30 climate districts in NSW
- information on the climate for each road link.

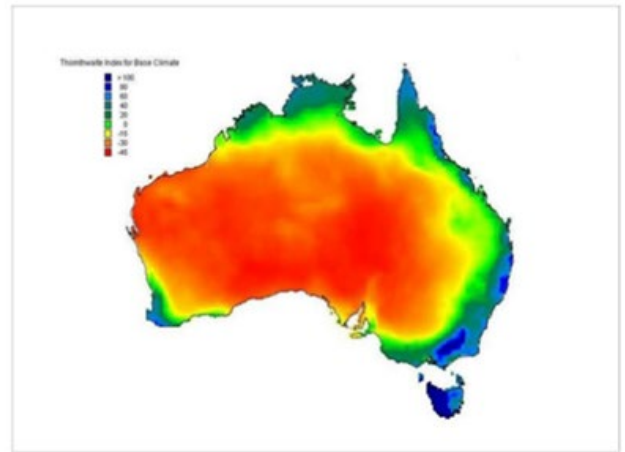
The supplied data was used to calculate Thornthwaite Moisture Index (TMI) (Thornthwaite 1948) for years 2014 to 2018 for the whole network using TMI calculation method 3 from a previous project (Austroads 2010c).

Figure 3.2 shows the distribution of TMI over the TfNSW road network (for year 2018) and its comparison against the TMI map for the whole of Australia. The calculated TMI values are drier than those observed in Austroads (2004), but the relative distribution is similar to those on the Australian TMI map.

Figure 3.2: Distribution of TMI for TfNSW road network (2018 values)



NSW TMI calculated and assigned to each road section using district climate information



TMI map of Australia in 2000 (Austroads 2004)

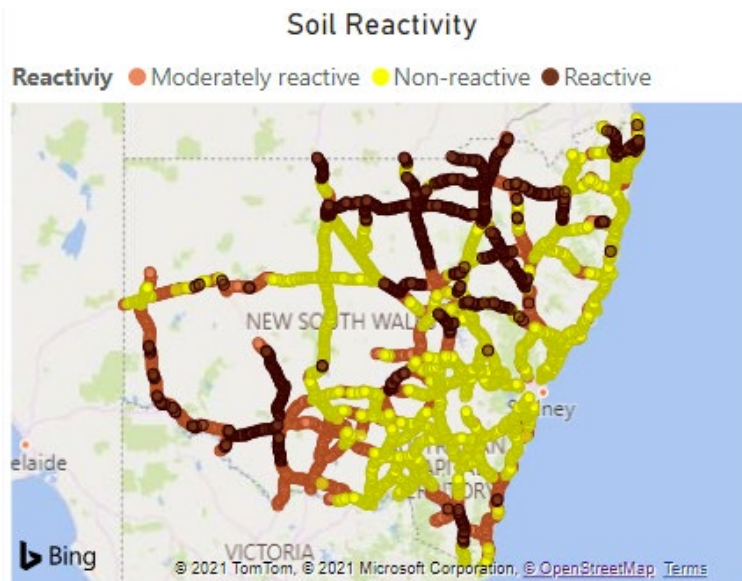
3.4.2 Soil Reactivity

The degree of soil reactivity determines the swelling and shrinkage behaviour of the soil with changes in moisture content. While sandy soils are generally non-reactive, a few types of clayey soils are highly reactive due to their smectite clay mineralogy. No direct information, in terms of the amount of subgrade soil reactivity, was available for any of the road networks. The data available from different public sources (CSIRO 2013, Viscarra Rossel 2014) was used to estimate the reactivity information based on clay content (%Clay, 0–30 mm) and clay mineralogy information. This was done as follows:

- The clay content map was compared with the clay mineralogy map (Viscarra Rossel 2014). It was found that the clay mineralogy of the subsoil lined up with the clay content with higher clay contents associated with the reactive clay minerals (montmorillonite/smectites).
- QGIS is a geographical information system (GIS) that has software allowing users to analyse and edit spatial information. In this application QGIS uses a point sampling technique to estimate clay content information on the TfNSW road segments.
- Assigning soil reactivity is based on the % clay content: < 30% is non-reactive; 30–45% is moderately reactive; and > 45% is reactive.

Figure 3.3 displays the estimated clay reactivity for the TfNSW network for 2018. Since the surveyed network varied from year to year, soil reactivity information was estimated for all segments surveyed in each year.

Figure 3.3: Soil classification based on reactivity for TfNSW road network (2018 values)



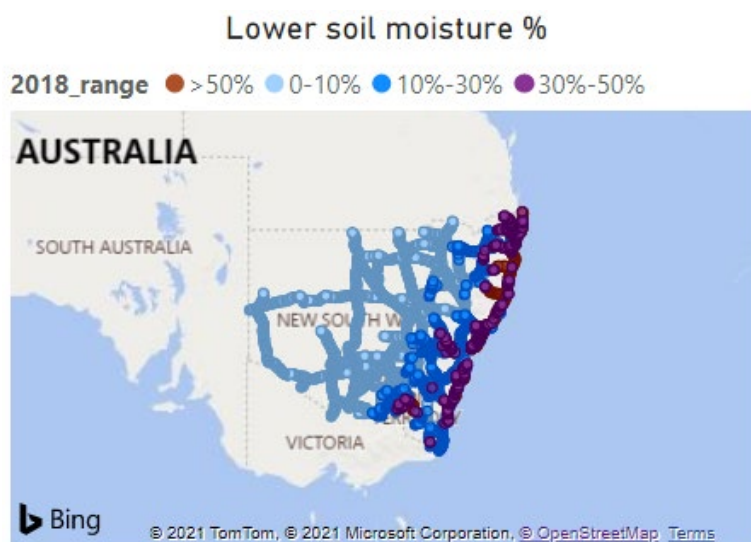
Sources: CSIRO (2013), Viscarra Rossel (2014).

3.4.3 Sub-soil Drainage Information

TfNSW was the only road agency able to provide terrain information in their initial data supply although no drainage condition data was supplied. In the absence of drainage information in NSW, the availability of the lower soil moisture (LSM) content percentage (%) was extracted for use as a surrogate to estimate drainage conditions. The historical LSM (2001–2021) percentage was extracted from the Bureau of Meteorology website (BOM 2021) for all of Australia.

Figure 3.4 displays the LSM percentage for the TfNSW network for 2018. Since the surveyed network varied from year to year, the LSM percentage information was estimated for all segments surveyed in each year.

Figure 3.4: Lower soil moisture (%) for TfNSW road network (2018 values)



Source: BOM (2021).

3.4.4 Additional Information Supplied by TfNSW

TfNSW provided the following additional information for the whole NSW network:

- Maintenance history – historical routine maintenance cost (\$/lane-km/year) from 2014 to 2021 for segments where some type of routine maintenance work was done.
- Asphalt thickness – estimated for the asphalt surfaced segments. The data specifies whether the thickness is more or less than 100 mm.
- Seal type information – provided for the sprayed seal surfaced segments. The data specified whether the seal type was single/single, double/double etc.

3.5 Review of the LTPP/LTPPM Data

LTPP data from both Australia and New Zealand was reviewed by the project. After the development of current Austroads RD models in 2010, a further 10 years of observational data was collected on the Austroads LTPP/LTPPM sites in the period from 2008 to 2018. The New Zealand NZTA LTPP program was established in 2001 with 176 sites providing 18 years of observational data. The proposed methodology primarily aimed to use the LTPP/LTPPM and NZTA LTPP datasets to update the current RD models.

3.5.1 Overview of Austroads LTPP/LTPPM Data

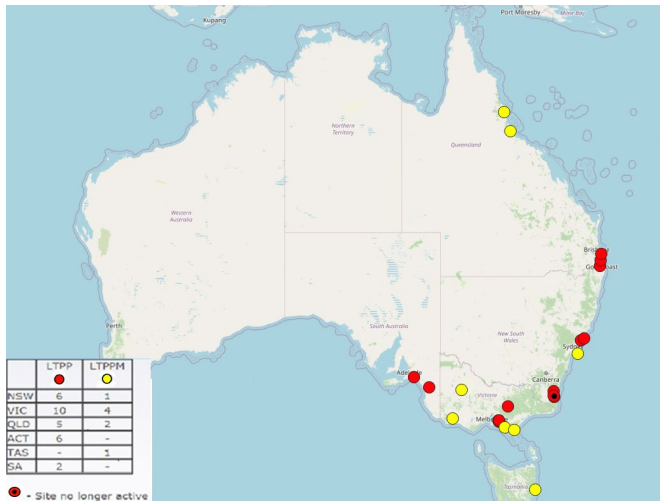
The Austroads LTPP/LTPPM program covered the following:

- 33 LTPP sites (39 sections)
- 8 LTPPM sites (40 sections)
- sites across all Australian jurisdictions except Western Australia and Northern Territory.

The Austroads LTPP/LTPPM database contained a detailed record of the following:

- section details including pavement composition
- roughness, rut depth, falling weight deflectometer (FWD) data
- maintenance history
- traffic
- environment, TMI.

Figure 3.5 shows the location of the LTPP and LTPPM sites across Australia, while Table 3.1 outlines features of the site groups and the number of sites and Table 3.2 summarises the observed performance of the site groups.

Figure 3.5: Austroads LTPP and LTPPM sites

Source: Austroads (2019b).

Table 3.1: Austroads LTPP and LTPPM sites

All pavement groups	No of sites	No of sections	Section length (m)	Test regime
LTPP	23	23	100–200	SHRP – LTPP
ACT-LTPP	6	12	800	Network survey
SA-LTPP	4	4	12	ARRB method
LTPPM	8	40	200	Reduced SHRP – LTPP

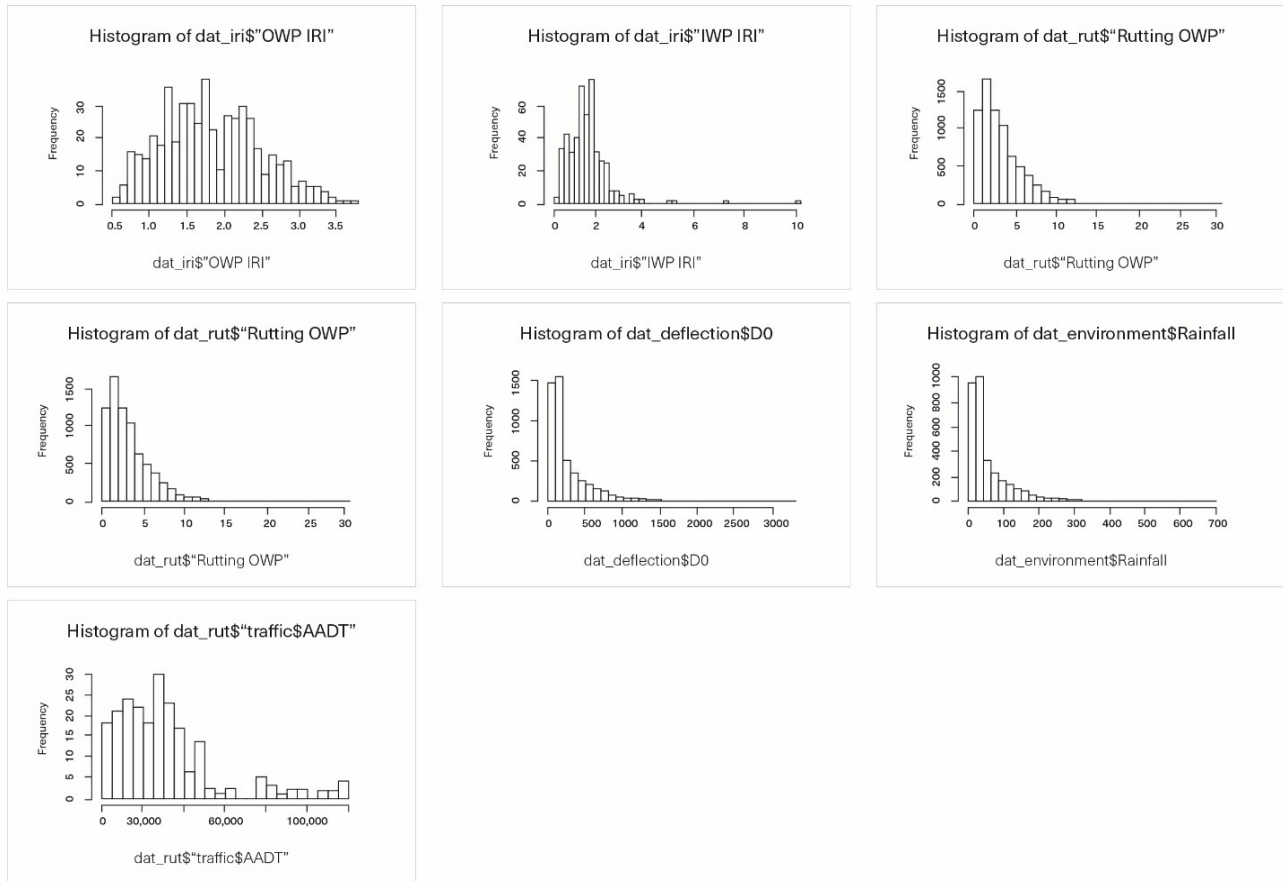
Source: Austroads (2019b).

Table 3.2: Condition progression rates for Austroads LTPP sites

Pavement group	Deflection (micron/yr)	Roughness (IRI/yr)	Rutting (mm/yr)	Cracking (%/yr)	Traffic (MESA/yr)
LTPP (SHRP+ALF)	11	0.07	0.20	1.92	0.4–8.8
LTPP (AAPA)	7	0.04	0.26	0.06	3.0–7.5
LTPP (rural highway)	72	0.03	0.78	0.06	0.02–0.8
ACT-LTPP	24	0.05	0.17	0.87	0.05–1.0

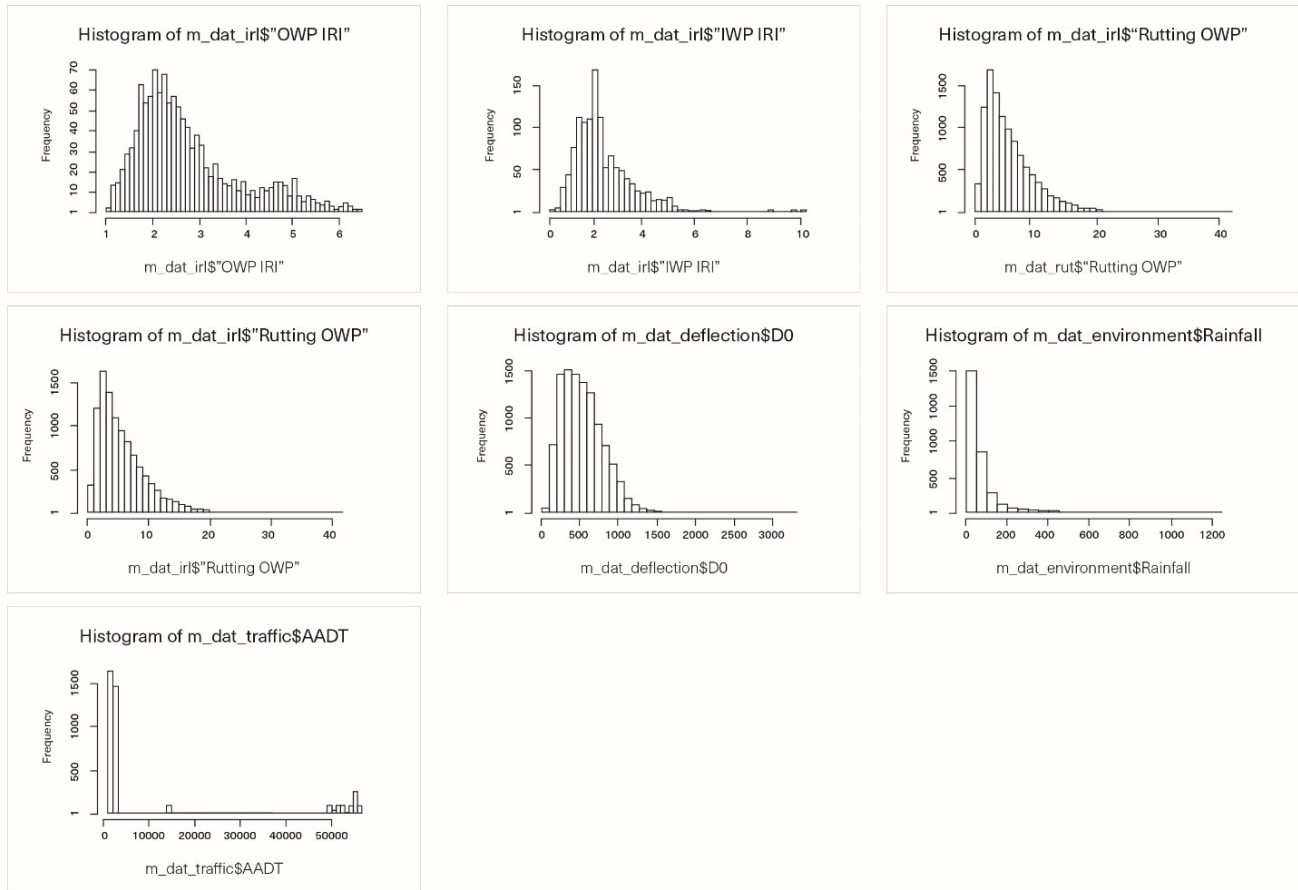
Source: Austroads (2019b).

Figure 3.6 and Figure 3.7 show the distribution of inner and outer wheel path roughness (IRI, m/km) and rutting (mm), maximum deflection (micron), rainfall (mm) and traffic (AADT) for the Austroads LTPP and LTPPM sites, respectively.

Figure 3.6: Distribution of various parameters – Austroads LTPP sites

The shapes of separate distributions of the parameters for the LTPP and LTPPM sites are generally similar, except for the traffic and rainfall distributions which are much narrower for the LTPPM sites, reflecting the relatively small number of 8 LTPPM locations compared to the 41 LTPP locations.

On the other hand, the maximum deflection distribution for the LTPP sites is narrower than that for the LTPPM sites, with a mean maximum deflection ($D0 = 250$ micron) and modified structural number ($SNC = 7.7$). The LTPP sites have a higher strength than the LTPPM sites that have a higher mean maximum deflection and lower strength ($D0 = 500$ micron; $SNC = 5.0$). The rutting and roughness distributions for the LTPPM sites have a greater range than the LTPP rutting and roughness distributions which also reflects the lower deflections (higher pavement strength) of the LTPP sites.

Figure 3.7: Distribution of various parameters – Austroads LTPPM sites

3.5.2 Usability of Austroads LTPP/LTPPM Data

The level of completeness in the Austroads LTPP/LTPPM datasets, including the availability of necessary additional information, such as traffic, environment, climate, maintenance history etc., make this an excellent basis for RD model update. However, the following data was not included in the LTPP/LTPPM datasets and required estimation as part of this data review and update process:

- soil reactivity
- drainage and terrain.

Soil reactivity

Soil reactivity for the Austroads LTPP/LTPPM sites was estimated using the same methods applied in Section 3.4.2 in estimating the soil reactivity for the TfNSW network.

Drainage and terrain

No terrain or drainage condition data was available for Austroads LTPP/LTPPM sites. In the absence of drainage information, the lower soil moisture content (LSM) data was extracted and used as a surrogate for drainage. This information was estimated using the same method applied in Section 3.4.3.

It was not possible to source terrain information for the LTPP/LTPPM sections, although most sections were located in relatively flat terrain.

3.5.3 Usability of New Zealand LTPP Data

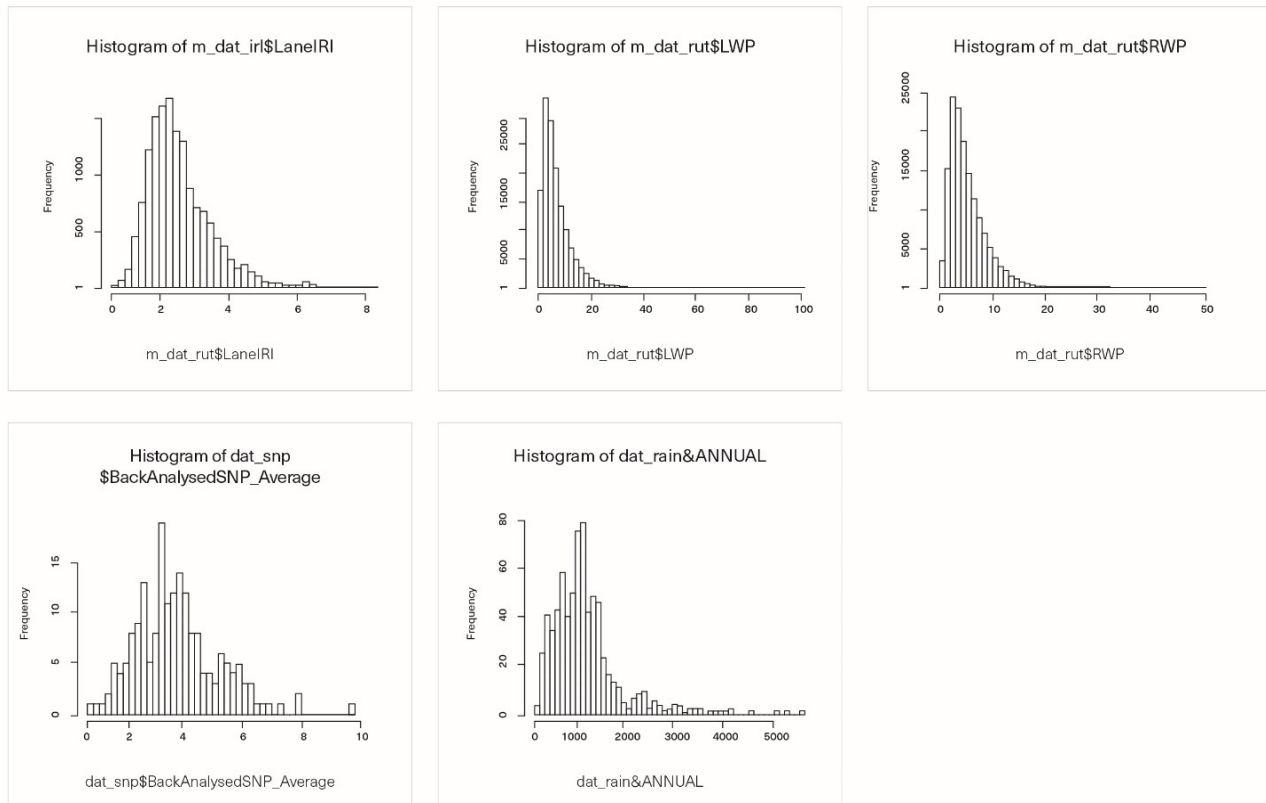
Some lack of completeness in the New Zealand LTPP dataset, including the availability of necessary additional information e.g. climate measured using TMI and drainage measured using LSM, made this dataset less suitable for the RD model update.

The Austroads and New Zealand LTPP datasets were compared using descriptive statistics and 't' tests in IBM SPSS Statistics software 2021. The results showed that:

- Roughness statistics (mean, median, range, etc.) were comparable between the Austroads and New Zealand LTPP sites.
- Rutting statistics for New Zealand LTPP sites were slightly higher than those observed on the Austroads LTPP sites.
- Deflection values were significantly different between the two datasets. The New Zealand LTPP sites displayed much higher mean deflections (lower strength) than the Austroads LTPP sites.

Figure 3.8 shows the distribution of inner and outer wheel path rutting (mm), lane roughness (IRI, m/km), adjusted modified structural number (SNP), and rainfall (mm) for the NZTA LTPP sites. The rainfall distribution is significantly greater than that for the Austroads LTPP/LTPPM sites, reflecting the larger number of NZTA LTPP sites (167). The NZTA LTPP sites approximate mean pavement strength, SNP, is around 3.5, significantly lower than the range of mean pavement strength of SNC from 5 to 7.7 for the Austroads LTPP/LTPPM sites.

The strength parameters, SNC and SNP are similar measures where SNP uses a weighting factor (Parkman & Rolt 1997) to SNC that reduces the contribution to pavement strength of the subbase and subgrade with their increasing depth. This weighting factor was introduced because the strength contribution of each layer in the pavement system is independent of depth with the SNC parameter resulting in over-estimates of SNC in the case of pavements greater than 700 mm thick.

Figure 3.8: Distribution of various parameters – NZTA LTPP sites

Due to the difficulty of estimating TMI and LSM variables in New Zealand and the fact that the New Zealand LTPP sites displayed much higher mean deflections than the Austroads LTPP sites, indicating substantial differences in the New Zealand subgrades compared to those in Australia, it was decided not to combine the Austroads and New Zealand LTPP datasets. It is also understood that New Zealand has highly resilient volcanic-derived soils which are atypical in comparison to Australian subgrades.

4. Development of the Cracking RD Model

4.1 General Approach and Limitations

As noted in Section 3.3.1, the cracking RD model was developed using the TSD cracking data provided by TfNSW. The cracking RD model was limited to thinly surfaced (sprayed seal and thin asphalt ≤ 40 mm) flexible unbound granular base pavements undergoing gradual deterioration. This was due to:

1. the general lack of cumulative cracking, $\sum \Delta crx$, observed on the wider ranging LTPP/LTPPM sites
2. cracking not being accurately and routinely measured until late in the LTPP/LTPPM monitoring program when the automatic crack detection (ACD) device was introduced
3. the majority of TfNSW TSD cracking data was for sprayed seal unbound granular pavements undergoing gradual deterioration.

The existing Austroads cracking model was based on cracking data from South Australia (Austroads 2010a). The historical annual cracking rates of progression in South Australia were observed to range from 0.16 to 4.15% cracking per year (Austroads 2008), while the LTPP/LTPPM dataset showed a range of cracking progression from 0.06 to 1.92% cracking per year (Austroads 2019b). The maximum level of cracking observed in South Australia (Austroads 2010a) was 53% while the maximum observed cracking from the TfNSW TSD cracking data was 18% (see Table 4.1) implying that all the TfNSW cracking data was undergoing gradual deterioration.

4.1.1 Data Summary

The complete Austroads LTPP/LTPPM database, including all sites with all years of collected data, was used at the beginning of the analysis to determine suitable data samples. However, not all sites could be used due to their pavement composition, condition progression, maintenance history etc. The following criteria were used to determine the suitability of the sites for analysis:

- Sites with bound/stabilised pavement bases were excluded due to different pavement deterioration characteristics compared with unbound granular pavements.
- Sites with thick asphalt surfaces (> 40 mm) were excluded.
- For the remaining LTPP/LTPPM sites, the condition data years that were free from the effect of the maintenance were retained.
- Further filtering of the above sites was done to only include years with an increasing trend in condition progression to exclude effects from unrecorded maintenance if any.

The above process significantly reduced the final sample size for analysis. The following tabulation summarises the characteristics of the TfNSW TSD cracking data, and the independent variables considered in the cracking RD model development.

Table 4.1: Summary of observational data used in the cracking RD model development

Data source	Independent variable range								Dependent variable range	No. samples
Variables	AGE ⁽¹⁾	MESA ⁽²⁾	TMI ⁽³⁾	Clay ⁽⁴⁾	SL ⁽⁵⁾	D _{0i} ⁽⁶⁾	LSMF ⁽⁷⁾	D ₀ ⁽⁸⁾	$\sum \Delta crx_i$ ⁽⁹⁾	
TfNSW	15–93	0.02–6.54	–48–12	0–18	–	14.5–1250	0.042–0.96	5.8–636	0–18	257

1. Pavement age (years) at time of observation.
2. Traffic load in millions of equivalent standard axles per lane per year.
3. Thornthwaite Moisture Index.
4. Clay (%) in subgrade.
5. Service life (years) increased SL related to increased cumulative ESAs.
6. Measured maximum deflection (micron).
7. Lower soil moisture.
8. Design maximum deflection (micron).
9. Cumulative cracking (%).

4.2 Treatment of Independent Variables

Section 3.4 shows how the independent variables TMI, Clay and LSM were derived for the TfNSW TSD dataset.

Two potential independent variables, MESA and AGE, were supplied by TfNSW for the cumulative cracking, $\sum \Delta crx$, model. Another possible independent variable, service life, SL, for the cumulative cracking, $\sum \Delta crx$, model was estimated based on the supplied value for MESA based on the approach used in previous Austroads (2010a) RD modelling. The SL estimates ranged from 40 to 60 years, increasing with the traffic capacity of the road. Estimation of the design deflection, D_0 , as a further possible independent structural variable is discussed in the next section.

4.3 Development of the Cracking RD Model

4.3.1 Design Deflection D_0 based on the D_0/D_{0i} Structural RD Model

The design deflection D_0 variable, or in-service initial deflection, D_0 (based on Benkelman Beam deflection), was considered as a possible independent variable for the cumulative cracking, $\sum \Delta crx$, model based on the D_0 dependent variable in the currently updated structural RD model. The D_0 variable was derived from measured D_{0i} values as in Equation 3 which were seasonally adjusted by the Austroads (2019a) approach.

$$D_0 = K_s \times D_{0i} \times [1 - AGE \times (0.012 + 0.001 \times MESA + 0.0000868 \times TMI)] \quad 3$$

where

K_s = structural calibration coefficient for adjustment to local conditions (default = 1.0)

AGE = pavement age (years)

TMI = Thornthwaite Moisture Index (1948) range –50 to 100)

MESA = annual millions of ESAs of traffic load per lane

4.3.2 Estimation of Cracking Age, $crxAGE$, of the TSD Cracking Data

The cracking age, $crxAGE$, is the number of years since cracking was initiated at the end of sprayed seal life. The cracking age, $crxAGE$, is estimated based on the age of the seal and the estimated time to initiate surface cracking as shown in Equation 4.

$$crxAGE = \text{seal age} - \text{seal life} \quad 4$$

The implication is that seal life coincides with, or is used as a surrogate for, the timing of crack initiation. Equation 5 estimates the seal life (Austroads 2010d) based on the time for the bitumen binder to harden, and therefore can be used for this purpose provided it is calibrated to local conditions.

$$\text{Seal life} = \left[\frac{0.158 \times TMIN - 0.107 \times R + 0.84}{0.0498 \times T - 0.0216 \times D - 0.000381 \times S^2} \right]^2 \quad 5$$

where

$TMIN$ = yearly mean of the daily minimum air temperature ($^{\circ}\text{C}$)

T = $(TMAX + TMIN)/2$

$TMAX$ = yearly mean of the daily maximum air temperature ($^{\circ}\text{C}$)

D = the ARRB Durability Test result (days, usually taken as 8 to 10)

S = nominal size of seal (nominal stone size, mm)

R = risk factor with a scale from 1 (very low risk) to 10 (very high risk)

The variables, $TMIN$ and $TMAX$, were available from the Bureau of Meteorology (BOM 2021) from meteorology stations near the sites of the data samples. The seal age of the data samples was included in the TSD cracking data supplied by TfNSW. Equation 5 assumes that all cracking was initiated by binder hardening. However, this is not always the case, as binders have additives that can change their performance such as polymer modified binders (PMBs).

Several other possible seal life estimates were also considered as the cracking age, $crxAGE$, is a critical variable in predicting cumulative cracking. This uncertainty was a significant source of error in the cracking RD model development. Another potential source of error was the reliability of the estimates of seal age which can suffer from a lack of accurate resealing records.

The risk factor, R , in Equation 6 is also somewhat arbitrary, however, Main Roads Western Australia (MRWA) has quantified R in terms of both rainfall and traffic as follows (MRWA 2019) which was considered for use in Equation 5.

$$R = 0.00097 \times \text{Rain} + 0.0064 \times (\text{AADT})^{0.5} + 0.00001 \times \text{Rain} \times (\text{AADT})^{0.5} \quad 6$$

where

Rain = annual rainfall (mm)

AADT = average annual daily traffic (vehicles/day)

In keeping with the original derivation of the seal life model (Equation 5) where seal life reflects an agency view of end of life, it was found that many samples already had cracking present. Therefore, applying Equation 5 directly was not considered useful as a predictor of crack initiation. It was then decided based on the experience of the team that where the surface age had reached 10 years this would be the age of crack initiation, or seal life. Only the samples which had a surface age greater than 10 years and were cracked and demonstrated an increasing trend of cumulative cracking over time were included in the cracking RD model dataset. However, it is recognised that the assumption applied may not reflect crack initiation timing throughout Australia and calibration of any model is essential.

4.4 Cumulative Cracking, $\sum \Delta \text{crx}$, RD Model Development

4.4.1 The Cracking RD Model

A total of 256 cumulative cracking samples, $\sum \Delta \text{crx}$, that were experiencing increasing cracking over the 4 years of TSD measurement were available for predicting sprayed sealed granular pavement cracking. The following model (Equation 7) was developed from this data by multi-variate non-linear regression (MVNLR) analyses by IBM SPSS Statistics software 2021.

$$\sum \Delta \text{crx} = K_c \times [\text{EXP}[0.227 \times \text{crxAGE} \times (100 + \text{TMI}) / 100] - 1] \quad 7$$

where

$\sum \Delta \text{crx}$ = cumulative cracking (% total lane area)

= crocodile cracking (%) + transverse cracking (%) + longitudinal cracking (%)

crxAGE = cracking age (elapsed time from the commencement of cracking, years), see Equation 4

TMI = Thornthwaite Moisture Index at time 'i'

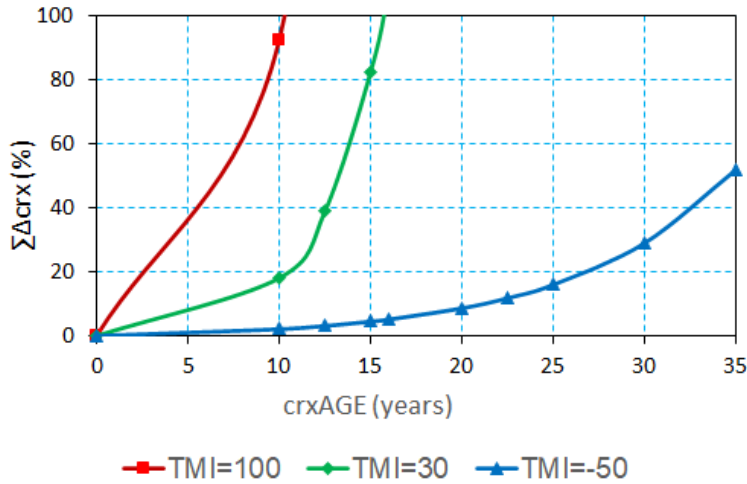
K_c = calibration coefficient for local conditions (default = 1.0)

The goodness of fit (r^2) of Equation 5 to the data was indeterminate due to the high variation in the observed cumulative cracking data. As noted in Section 4.3.2, this was largely attributed to not being able to reliably determine the seal life on each of the samples. However, despite the lack of data fit, the crxAGE and TMI were the two independent variables that were statistically significant. Appendix A provides the outputs of this statistical analysis, although they are limited due to the lack of fit to the data and there being only two statistically significant variables.

None of the other independent variables examined, MESA, D_0 , %Clay and lower soil moisture content factor, LSMF, were statistically significant. The %Clay and LSMF independent variables were correlated with TMI and therefore were not included.

Figure 4.1 shows the major variation in the rate of cumulative cracking, $\sum \Delta \text{crx}$, due to the wide range of TMI values with cracking age, crxAGE.

Figure 4.1: Prediction of cumulative cracking



This model predicts that under a high TMI of 100, the sprayed seal will reach 100% cracking after 10 years, while for a lower TMI of 30, the sprayed seal will reach 100% cracking after 15 years. Under the lowest possible TMI of -50, the model predicts over 50% cracking after 35 years.

4.4.2 Annual Incremental Cracking

The annual increase in incremental cracking, $\Delta crx(t)$, is simply found by subtracting the cumulative cracking at time 'i + 1', $\Delta crx(t)_{i+1}$, from the cumulative cracking at time 'i', $\Delta crx(t)_i$, provided the cracking time, crxAGE is known. This approach is expressed as shown in Equation 8 using Equation 7.

$$\Delta crx(t) = K_c \times \{ [\text{EXP}[0.227 \times crxAGE_{i+1} \times (100 + TMI)/100] - 1] - [\text{EXP}[0.227 \times crxAGE_i \times (100 + TMI)/100] - 1] \} \quad 8$$

where all terms are as previously defined.

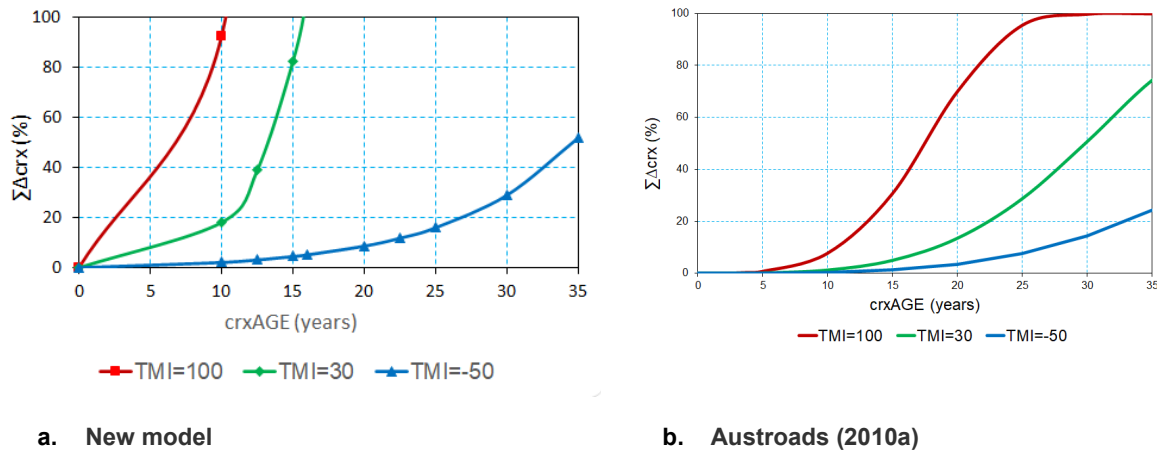
4.4.3 Comparison with Austroads 2010 Cracking RD Model

A direct comparison between the cumulative cracking model developed here and the existing Austroads (2010a) cumulative cracking model for sprayed seals can be made, even though the model forms are different, but meet the boundary conditions. The Austroads cumulative cracking RD model is as shown in Equation 9 which predicts an 's' shape for cracking progression.

$$\Sigma \Delta crx = K_c \times [100 - 200 \times (1 + \text{EXP}((0.234 \times crxAGE / ((200 - TMI)/25))^{3.5}))^{-1}] \quad 9$$

where all the variables are as previously defined.

Figure 4.2a and b compare the predictions of cumulative cracking using the new upgraded cracking RD model against the existing Austroads cracking RD model.

Figure 4.2: $\Sigma\Delta crx$ vs $crxAGE$ 

Apart from the different functional model forms for the prediction of cracking progression, the times to reach 100% cracking for the various values of TMI are different with the Austroads (2010a) predicting much slower rates of crack progression for all values of TMI.

This variation can be attributed to some extent to the location of the samples used to develop the cracking models. The Austroads (2010b) cracking model used cracking data from South Australia which comprised 1675 samples where the extent of cracking was up to 53%. The new cracking model was based on the TfNSW TSD cracking data collected over 4,000 km of arterial road length with time series data for five years covering 11% of the NSW sealed arterial road network (see Section 3.3.1). In both cases the model fit to the data were poor to non-existent.

The cracking RD model will need calibration, i.e. adjustment of K_c , to suit local conditions as the current evidence suggests that location has a bearing on the rates of cracking progression.

Neither of the above cumulative cracking models include traffic, MESA, and pavement design deflection, D_0 , implying that thin surface cracking is not related to traffic and pavement strength. This infers that the pavement is designed to match the expected traffic and the cracking is therefore not due to any structural or pavement material deficiency.

4.4.4 Potential for Further Refinement of Cumulative Cracking Prediction

Large amounts of TSD cracking data will be forthcoming in the future years from most jurisdictions in Australia. This will provide a large dataset of continuous cracking data, based on ACD measurement, over several years once the non-cracked road segments are filtered out.

As noted in Section 4.3.2, the estimation of cracking age, $crxAGE$ or seal life, is a critical variable in predicting cumulative cracking and is a major source of error in the cracking RD model development if the cracking age is inaccurate. Further refinement of sprayed seal life models needs to be undertaken as well as being able to accurately determine the types of bitumen and additives used in the binders being assessed for cracking by the TSD. This will allow application of the most appropriate seal life models, when available, to determine with a reasonable degree of reliability the sprayed seal life and hence the cracking age.

However, there is an alternative probabilistic approach to estimating crack initiation i.e. cracking age, used by Henning and Brown (2016) that is purely statistical based on multivariate logistic regression using seal samples that are uncracked and seal samples that are lightly cracked (< 10% cracked). This approach could use several additional variables such as stone size, traffic, rainfall, and TMI, excluding the binder properties.

However, refinement of the cracking RD models should not be undertaken until several more years of TSD cracking data become available across most states and territories in conjunction with reliable sprayed seal life models. Models are currently being investigated by Queensland Department of Transport and Main Road's (TMR) National Asset Centre of Excellence (NACOE) research program (Noya & Hwayyis 2021), and potentially by other agencies.

5. Conclusions

A cracking RD model using cumulative cracking, $\sum \Delta crx$, as the dependent variable for thinly surfaced flexible unbound pavements, was developed using the New South Wales dataset collected by the TSD from 2014 to 2018. The model has two significant independent variables: cracking age, $crxAGE$, and the Thornthwaite Moisture Index, TMI. Although the independent variables are statistically significant, the model has an indeterminate fit (r^2) to the data. Consequently, the cracking RD model will need calibration to suit locally observed cracking conditions as the evidence suggests that location also has a bearing on the rates of cracking progression.

As the cracking RD model development was highly dependent on the accuracy of the estimation of sprayed seal life and knowledge of the sprayed seal age, further refinement of sprayed seal life models needs to be undertaken as well as the supply of more reliable information on the types of bitumen and additives used in the binders being assessed for cracking by the TSD. This will allow application of the most appropriate seal life models, when available, to determine with a reasonable degree of reliability the sprayed seal life and hence the cracking age.

An alternative probabilistic approach to estimating crack initiation used by Henning and Brown (2016) could be considered based on multivariate logistic regression using seal samples that are uncracked and lightly cracked. Forecasting a discrete event such as crack initiation is best simulated using a probabilistic model. This approach could be considered in conjunction with the TMR's NACOE research program which is aimed at developing reliable sprayed seal life models.

However, refinement of the cracking RD models should not be undertaken until several more years of TSD cracking data become available across most states and territories in conjunction with reliable sprayed seal life models.

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Appendix A Multivariate Non-Linear Regression Analysis Outputs

A.1 Cumulative Cracking, $\sum \Delta \text{crx}$, RD Model

Nonlinear Regression Analysis

Notes

Output created	19-JUL-2022 12:07:40	
Comments		
Input	Active dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split file	<none>
	No.?? of rows in working data file	256
Missing value handling	Definition of missing	User-defined missing values are treated as missing.
	Cases used	Statistics are based on cases with no missing values for any variable used. Predicted values are calculated for cases with missing values on the dependent variable.
Syntax	MODEL PROGRAM a1=0.5. COMPUTE PRED_=EXP(a1*Age_Cracking*(100 + TMI)/100) - 1 . NLR Cracking_of_total_lane_area /OUTFILE='C:\Users\ranita.sen\AppData\Local\Temp\spss38652\SPSSFNLR.TMP' /PRED PRED_ /CRITERIA SS CONVERGENCE 1E-8 PCON 1E-8.	
Resources	Processor time	00:00:00.02
	Elapsed time	00:00:00.02
Files Saved	Parameter estimates file	C:\Users\ranita.sen\AppData\Local\Temp\spss38652\SPSSFNLR.TMP

Iteration History^b

Iteration number ^a	Residual sum of squares	Parameter a1
1.0	19065.427	.500
1.1	3299.734	.375
2.0	3299.734	.375
2.1	1486.207	.273
3.0	1486.207	.273
3.1	1392.655	.228
4.0	1392.655	.228
4.1	1392.638	.227
5.0	1392.638	.227
5.1	1392.638	.227
6.0	1392.638	.227
6.1	1392.638	.227

Derivatives are calculated numerically.

a. Major iteration number is displayed to the left of the decimal, and minor iteration number is to the right of the decimal.

b. Run stopped after 12 model evaluations and 6 derivative evaluations because the relative reduction between successive residual sums of squares is at most $SSCON = 1.00E-008$.

Parameter Estimates

Parameter	Estimate	Std. error	95% Confidence interval	
			Lower bound	Upper bound
a1	.227	.014	.201	.254

ANOVA^a

Source	Sum of squares	df	Mean Squares
Regression	594.710	1	594.710
Residual	1392.638	255	5.461
Uncorrected Total	1987.347	256	
Corrected Total	1317.200	255	

Dependent variable: Cracking(%_of_total_lane_area)^a

a. $R^2 = 1 - (\text{Residual sum of squares}) / (\text{Corrected sum of squares}) = ..$



Austroads

Level 9, 570 George Street
Sydney NSW 2000 Australia

Phone: +61 2 8265 3300

austroads@austroads.com.au
www.austroads.com.au